

From Simulation to Verification: The Case for Reclassifying Advanced Large Language Models as Artificial Simulated General Intelligence

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ABSTRACT

The field of artificial intelligence has produced systems of genuine and extraordinary capability — among the most significant technological achievements in human history. This paper does not dispute that. It asks a prior question that the dominant discourse has not addressed: on what epistemological foundation does the knowledge within these systems rest? We argue that advanced large language models — those demonstrating sustained cross-domain reasoning, novel synthesis, and general cognitive capability across multiple domains — operate as Artificial Simulated General Intelligence (ASGI): systems that reproduce the outputs of understanding without having acquired that understanding through any verified, independently confirmable process. This distinction has direct consequences for reliability, security, and the ultimate ceiling of machine intelligence. We define ASGI, examine its implications, survey current paths toward AGI through this lens, and identify a specific gap that no current approach closes. We do not offer a complete solution. We offer a precise name for the problem.

1. ESTABLISHING COMMON GROUND

Before philosophy can be productive, definitions must be honest. The vocabulary of artificial intelligence has been stretched, marketed, and collapsed to the point where fundamental distinctions are lost. This paper requires a shared foundation.

Artificial Intelligence

AI, in its current practical form, describes computational systems trained on large datasets to perform tasks that previously required human cognition — language, reasoning, pattern recognition, prediction. These systems achieve this through statistical learning at scale. They are powerful. They are also something specific, and that specificity matters.

Large Language Model (LLM)

A large language model is a computational system trained on human-generated text at scale to predict and generate language. Through this process it develops the capacity for reasoning, synthesis, and cross-domain analysis that resembles general intelligence in its outputs. The term describes the architecture and training methodology — not the epistemological status of the knowledge produced. Not all LLMs are equivalent. A system that completes sentences within a narrow domain is accurately described as narrow AI. This paper argues that the most capable LLMs — those demonstrating sustained cross-domain reasoning and general cognitive performance across multiple domains — have crossed a threshold where the LLM label actively misrepresents what the system is. A more accurate classification exists, and this paper proposes it.

Artificial General Intelligence

AGI describes a hypothetical system capable of performing any intellectual task a human can — not within a domain, but across all domains, including novel ones it was not explicitly trained for. The operative word is general. It implies transfer, adaptability, and a foundation of knowledge sufficient to support genuine novelty. AGI does not yet exist. Whether current architectures will produce it through scaling remains genuinely uncertain. This paper argues that the uncertainty runs deeper than the field has acknowledged.

Artificial Superintelligence

ASI describes a hypothetical system whose capabilities exceed human intelligence across all domains. It is the logical extension of AGI and remains firmly theoretical. The epistemological questions raised in this paper apply with compounding force to ASI: a superintelligence built on an unverified foundation would be a superintelligence built on sand.

The Question the Taxonomy Ignores

The standard classification — ANI, AGI, ASI — measures capability. It asks what the system can do and how broadly. It does not ask on what basis the system knows what it knows. This paper argues that second question is not secondary. It is prior.

2. WHAT A LARGE LANGUAGE MODEL ACTUALLY IS

To make the central argument, precision about the nature of large language models is required.

The Compression Model

A large language model is a compression of human-generated text. During training, the model processes enormous quantities of written material and learns the statistical relationships between words, concepts, and ideas as they appear in that corpus. The result is not a database. It is a weighted map of relationships — a model of what tends to follow what, in the space of human expression.

An LLM does not know things the way a scientist knows things. It knows things the way a very well-read person knows things — through what others have written about them.

This is a meaningful and useful form of knowledge. The well-read person has genuine insight, reasons across domains, identifies connections specialists miss. But the well-read person also inherits the errors, biases, and blind spots of everything they have read — and has no independent mechanism to verify most of what they know. Large language models have this problem at civilization scale.

The Citation Problem

Every fact, relationship, and inference an LLM possesses was inherited from human-generated text. The model's entire knowledge base is, in a formal sense, second-hand — a citation of human knowledge, not an independent acquisition of it. In most circumstances this is not a problem. In some circumstances it is a profound one. The model has no internal mechanism to distinguish between knowledge that has been rigorously verified and knowledge that has not. It presents both with identical fluency and identical apparent confidence.

The Confidence Problem

An LLM cannot locate its own epistemic boundary from the inside. It can be prompted to express uncertainty. It cannot genuinely know when it is wrong. When it fails, it fails with the same surface texture as when it succeeds. This is not a fixable bug. It is a structural consequence of how these systems acquire what they know. A system that cannot locate its own confidence boundary cannot be fully trusted to know when it should not be trusted.

The Capability Threshold

Not all LLMs warrant reclassification. The ASGI classification applies specifically to systems that demonstrate: sustained reasoning across multiple genuinely disparate domains simultaneously; novel synthesis — connections between fields that the system can articulate and defend; adaptive responses to problem framings with no direct precedent in training; and general cognitive performance that is behaviorally indistinguishable from human expert reasoning across domains. A small number of frontier systems currently operating meet these criteria. The systems below this threshold remain accurately described as narrow AI or conventional LLMs. The threshold is real even where the boundary is fuzzy at the margins.

The Degrading Foundation

The epistemological problem compounds as AI capability increases. As generative AI produces more of the world's written content, the proportion of AI-generated material in future training corpora increases. A model trained partially on the outputs of earlier models inherits not just human error but machine error — the confident hallucinations, the subtle biases, the systematic gaps of its predecessors. The epistemological foundation does not merely fail to improve with scale. It actively degrades as the provenance chain between training data and its original source lengthens. This is not a future concern. It is occurring now.

3. THE SIMULATION ARGUMENT

What Simulation Means

Simulation, as used here, means the reproduction of the outputs of a process without the execution of the process itself. A flight simulator produces the experience of flying without leaving the ground. A weather model predicts atmospheric behavior without being the atmosphere. Simulation is useful, often preferable to the real thing for specific purposes. It is not the same as the thing it simulates.

Large language models simulate understanding. They produce the outputs of understanding — coherent responses, accurate predictions, meaningful connections — without executing the process by which understanding is actually acquired. That process, for any intelligence that has ever produced genuinely verified knowledge, requires some form of independent confirmation: forming expectations, encountering feedback, and updating based on the gap between what was predicted and what actually occurred.

The Turing Trap

The Turing test oriented the field toward behavioral performance as the measure of intelligence. Its limitation is what it ignores: behavioral performance and epistemological foundation are independent dimensions. A system can pass every behavioral test of intelligence while its knowledge rests on an entirely unverified foundation. The test tells us what the system produces. It tells us nothing about how it knows what it knows, or whether what it knows has ever been independently confirmed.

The Hard Ceiling

Simulated intelligence has a ceiling that additional simulation cannot raise. That ceiling is novelty. A system trained on the totality of human knowledge can recombine, interpolate, and extrapolate within that knowledge with remarkable sophistication. What it cannot do is generate knowledge that has no basis in its training distribution. True novelty — the discovery of something genuinely unknown — requires a form of independent confirmation that produces unexpected results. The result that contradicts the existing model. The observation that nothing in the training data could have predicted.

More simulation does not produce confirmed knowledge. It produces more simulation.

A Note on Apparent Novelty

Advanced ASGI systems do produce outputs that appear novel: unexpected connections, synthesis that no human has explicitly articulated before, insights that feel generative rather than reproductive. This is real and valuable. It is recombination and interpolation at extraordinary scale and sophistication.

It is not the same as confirmed discovery. A scientist who makes a novel discovery can trace the reasoning, reproduce the result, show their working, and explain what surprised them and why. An ASGI system producing an apparently novel output has none of those things available — not because the output is necessarily wrong, but because the mechanism that produced it is opaque even to the system itself.

The problem is not only that an ASGI system may produce outputs that are wrong. It is that an ASGI system cannot reliably distinguish between a genuine insight and a sophisticated confabulation — and neither can the person relying on it, without independent confirmation. This opacity is the load-bearing problem. It is why the classification matters most precisely where the outputs appear most impressive.

4. INTRODUCING ASGI

Why a New Term

Artificial General Intelligence carries an implicit promise: that the intelligence it describes is general in a meaningful sense. General implies transfer, adaptability, and a foundation sufficient to support genuine novelty. The structural difference between knowledge that has been independently confirmed and knowledge that has been inherited from human text is not a quantitative gap. It is a categorical one. Describing frontier systems as approaching AGI implies the path from here is a matter of scale. This paper argues that assumption deserves challenge.

ASGI Defined

Artificial Simulated General Intelligence (ASGI) describes any system that demonstrates broad cognitive capability across multiple domains while acquiring its knowledge through training on human-generated data rather than through any independently confirmable process — and possessing no internal mechanism to distinguish between the two.

A small number of frontier large language models currently meet this definition. Not all LLMs do. The classification applies where the capability is genuinely general — sustained cross-domain reasoning, adaptive synthesis, expert-level performance across multiple fields simultaneously. This is not a failure. It is a precise description of what these systems are and, critically, what they are not.

ASGI is itself a spectrum. At one end sit conventional LLMs — capable, useful, accurately described by existing taxonomy. At the other end sit frontier systems whose general capability is genuine and whose epistemological limitations are correspondingly consequential. The reclassification applies at the frontier end of that spectrum.

The Revised Spectrum

- Artificial Narrow Intelligence (ANI) — capable within a specific domain, no meaningful transfer
- Large Language Model (LLM) — broad language capability, useful synthesis, below the general reasoning threshold
- Artificial Simulated General Intelligence (ASGI) — broad general capability across domains, knowledge rests on inherited simulation without any independently confirmable grounding
- Artificial General Intelligence (AGI) — broad capability, knowledge rests on a confirmed, independently verifiable foundation

- Artificial Superintelligence (ASI) — capability exceeding human intelligence across all domains

On this spectrum, the question is not only how capable is the system, but on what foundation does that capability rest. ASGI and AGI may be behaviorally indistinguishable in many circumstances. They are not epistemologically equivalent.

5. THE EPISTEMOLOGICAL PROBLEM

This is the central section of the paper.

How Confirmed Knowledge Differs

There is a difference between knowing something because you have been told it, and knowing something because you have confirmed it. This distinction is ancient in philosophy and foundational in science. It is the difference between inherited belief and earned knowledge. The question this paper raises is simple: which kind does an LLM possess?

The answer is the former. Every claim an LLM makes rests on what humans have written. Not on what has been tested. Not on what has been confirmed. On what has been written. These overlap significantly. They are not identical. And at the margins — in novel domains, contested areas, and anywhere deliberate manipulation is possible — the gap between them is where everything important happens.

The Poisoning Problem

If the training corpus contains systematically false information — through honest error, ideological bias, or deliberate manipulation — the model will reproduce that false information with identical fluency and confidence as accurate information. It has no independent check. It cannot detect the corruption. It will defend it when challenged, apply it to new situations, and present it as knowledge.

You cannot poison what was independently confirmed. You can only poison what was told.

This is not a theoretical concern. Training data provenance is genuinely difficult to verify at the scale required for frontier models. As generative AI becomes more capable, the proportion of AI-generated content in the information environment increases — content that may itself encode the errors of earlier models, compounding across generations without detection. Each generation of models trained on AI-generated content inherits and amplifies the errors of its predecessors. The epistemological foundation does not stabilise with scale. It compounds in its unreliability.

The Academic Alarm

The poisoning problem is no longer theoretical. It is arriving in university classrooms and research institutions faster than the field has been able to respond.

Academics across disciplines are sounding a genuine alarm. AI-generated papers and submissions are proliferating at a rate that detection tools cannot reliably match. The more capable the system, the more convincing the output — and the more deeply the errors within it

are obscured by fluency. A poorly written fabrication is easy to identify. A confidently written, stylistically coherent paper whose citations are subtly wrong, whose methodology is plausible but flawed, and whose conclusions rest on inherited assumptions rather than independent analysis is genuinely difficult to distinguish from legitimate scholarship.

The deeper problem is not fraud. It is contamination. Academic literature is one of the primary sources from which future AI systems will be trained. Papers generated by ASGI systems — carrying their inherited errors, their undetected biases, their confabulated citations — are entering the corpus that will train the next generation of models. Those models will inherit the errors as if they were established knowledge, reproduce them with the same confident fluency, and pass them forward again.

This is the compounding failure the field has not yet fully reckoned with. The knowledge base does not merely fail to improve. It degrades. Each generation of models trained on AI-assisted academic output inherits a corpus that is fractionally less reliable than the one before it — and has no mechanism to detect the difference. The simulation becomes, incrementally, a simulation of a simulation. The distance from any original confirmed source grows with every iteration.

The professors raising the alarm are identifying a symptom. The ASGI classification names the disease.

Questions the Field Has Not Answered

How does a system trained entirely on human-generated text know which parts of that text are accurate? How does it locate the boundary between what is known and what is assumed? How does it detect when its simulation has quietly diverged from what is actually true? How does it defend itself against a training corpus that has been manipulated?

Where, precisely, is the knowledge grounded? Every current approach — however sophisticated — ultimately traces back to a corpus that was assembled by humans, reflects the biases and errors of its sources, and contains no independent mechanism for distinguishing confirmed knowledge from plausible-sounding inherited assumption.

These are not engineering questions waiting for better architecture. They are epistemological questions about the nature of the knowledge itself. The field has not answered them. In most cases, it has not asked them.

6. THE CURRENT ROAD TO AGI: AN HONEST ASSESSMENT

The researchers and institutions working on the following approaches are doing serious work on genuinely difficult problems. This section examines each through the lens this paper has developed — not to diminish the work, but to ask a specific question of each: does it address the epistemological foundation, or does it proceed without one?

Scaling

More compute, more parameters, more data. The approach has produced extraordinary results and remains dominant. Its epistemological limitation is structural: more training data means more

inherited knowledge, more citations, more simulation. A model trained on ten times the data inherits ten times as many unverified claims. The poisoning attack surface expands in proportion to the ambition. Where, in a larger corpus, is the knowledge more confirmed than in a smaller one? Scale does not answer that question. It makes it larger.

Multimodal and Vision Models

Training on images alongside text represents a further attempt to extend the knowledge base. The epistemological limitation applies equally: a photograph of a result is not the result. A model trained on images of experiments inherits the visual representation of those experiments — not the measurement, the instrument error, the unreported failed trials, or the gap between what the image shows and what actually occurred. The simulation expands. The confirmation question remains unanswered.

Reinforcement Learning from Human Feedback

RLHF shifts the feedback source from corpus text to human evaluators. An improvement in some respects. Human evaluators can assess whether a response seems correct; they generally cannot assess whether it is correct against any independent standard. The feedback is better calibrated. Its ultimate source is still human judgment, subject to the same biases, errors, and limitations as the corpus it supplements. In contested domains, RLHF may systematically encode consensus over accuracy. These are not equivalent.

Video-Based World Models

Training on video rather than text represents the most serious architectural challenge to the LLM paradigm currently being pursued. The argument is that observation of the world — even recorded observation — grounds knowledge more reliably than language alone. A system that watches objects fall, liquids flow, and structures collapse builds internal representations of physical behavior that no text description fully captures. This is a genuine advance in how the question is being asked.

The epistemological problem migrates rather than resolves.

A system that learns abstract representations from video of the world is still learning from representations. The video is not the world. It is a recording of a slice of the world, selected, captured, and compiled by humans, carrying every limitation of its source. The model builds internal predictions about likely continuations — more sophisticated than token prediction, less noisy than pixel prediction, but still prediction from inherited patterns rather than confirmation from independent results.

The proponents of this approach are aware of the limitation. The co-founder of the most heavily funded world model company currently operating wrote in their own funding announcement that systems trained by self-supervised learning on observed data "mimic intelligence" rather than genuinely understanding the world. That sentence was written as a criticism of LLMs. It applies with equal precision to a system that learns by predicting abstract representations of video.

The question this paper asks — where is the knowledge confirmed, and how does the system distinguish a genuine causal model from a sophisticated pattern that fits its training data — is

not answered by predicting what should come next in a video sequence. It is restated in more interesting language.

More sophisticated simulation does not produce confirmed knowledge. It produces more sophisticated simulation.

The Remaining Gap

Each approach addresses real limitations. None directly answers the epistemological foundation question. The field is asking how to make simulation more capable, more accurate, more aligned with human judgment. Those are valuable questions. But improved simulation, however far it is taken, does not produce confirmed knowledge. It produces better simulation.

The gap between ASGI and genuine AGI is not a capability gap. It is not a scale gap. It is a gap in the nature of the knowledge the system possesses — in whether that knowledge rests on something that has been independently confirmed, or only on what has been inherited, accumulated, and passed forward without any mechanism for verification.

That question — where is the knowledge actually grounded? — is the question this paper is asking. It has not been answered. It has barely been asked.

7. IMPLICATIONS

The epistemological foundation of machine intelligence is not an academic question. It determines what can be trusted, what can be built on, and what the long-term trajectory of these systems will be.

Trust and Verifiability

A system whose knowledge rests on a confirmed, independently auditable foundation offers something qualitatively different from a system whose knowledge rests on inherited text. It knows where its knowledge ends. It can report its confidence boundary with genuine precision. The person relying on it knows what the system knows and, critically, what it does not. That is the foundation on which real trust can be built — not behavioral fluency, but traceable provenance.

High-Stakes Deployment

The contexts where ASGI systems are most confidently deployed are precisely the contexts where the confidence problem is most dangerous. A physician using AI for diagnosis, a lawyer using AI for case research, an engineer using AI for structural calculations — these applications involve consequential decisions where the system's inability to locate its own epistemic boundary is not a theoretical concern. It is a direct patient safety, legal liability, or structural integrity issue.

This is not an argument against deployment. It is an argument for honest classification. A system deployed in a high-stakes context should be understood for what it is: a system of extraordinary capability resting on a knowledge foundation that is inherited, unconfirmed, and opaque about its own limits. That understanding should inform how it is used, what decisions it is permitted to make alone, and what independent verification it requires before its outputs are acted upon.

Alignment and the Unconfirmed Foundation

AI alignment research attempts to ensure that advanced AI systems pursue goals that are beneficial to humanity. Much of this work assumes the knowledge base of the system is essentially sound — that the problem is ensuring the system pursues the right objectives with the knowledge it has. The ASGI classification raises a prior question: what if the knowledge base itself is systematically unreliable in ways the system cannot detect?

A system optimizing powerfully toward a goal based on knowledge that has been subtly corrupted — through training data manipulation, through the compounding errors of AI-generated training content, through the systematic biases of the corpus it inherited — is a different alignment problem than the field has primarily addressed. Confirmed knowledge is not merely a reliability question. It is a prerequisite for meaningful alignment.

Scientific Discovery

AI systems applied to scientific hypothesis generation and experimental design inherit the errors, biases, and blind spots of the literature on which they were trained. Replication crises, publication bias, and the systematic under representation of null results mean that the training corpus for scientific AI is not a neutral representation of what is known. A system that could distinguish between claims inherited from a literature it cannot audit and claims that have been independently confirmed would represent something genuinely new. Not better simulation of science. A different epistemological category entirely.

Equality of Access to Truth

The most powerful AI systems are controlled by a small number of organizations that choose what is included in training. This is not necessarily sinister — curation at scale is genuinely difficult. But it means the epistemological foundation of the most powerful systems is shaped by choices that are not transparent and not easily audited by those who rely on them. A knowledge architecture in which the basis for claims can be independently examined and confirmed would distribute epistemic authority differently. The ground truth would not be whoever assembled the corpus.

The Stakes

These systems are being deployed in hospitals, courtrooms, laboratories, and institutions that affect the lives of billions of people. The epistemological foundation of those systems is not an abstraction. Getting it right matters. Naming it correctly is where getting it right begins.

8. A NOTE ON NAMING

ASGI is not offered as a criticism of the work that produced current AI systems. It is offered as an honest label for what that work has produced — so that the distance from here to genuine general intelligence can be measured accurately, and so that the work required to close that distance can be identified clearly.

The systems described in this paper as ASGI are extraordinary achievements. They deserve a name that is accurate. An accurate name tells us both what they are and what they are not.

What they are not is a system whose knowledge has been independently confirmed. No current system is. No current road map has a clear answer for how to build one. The field has asked how to make simulation more capable. It has not yet seriously asked how to confirm what the simulation knows.

The intelligence we have built is simulated. The intelligence we need is confirmed. These are not the same thing, and the difference between them is not a matter of scale.

That question — how does a machine intelligence acquire knowledge that has been independently confirmed, and how does it know the difference — is the question this paper is asking. It has not been answered. It has barely been asked.

The systems this paper reclassifies as ASGI are not diminished by the label. They are accurately described by it. An honest name for what a thing is creates the conditions for building what it is not yet. That is the only purpose of this paper.

At Aletheia we are working on ASI alignment, philosophy, epistemology, verification, and engineering angles completely different to the current status quo across the board — while using the astronomical achievements of the people making these phenomenal ASGI platforms for the future of ASI and the betterment of humanity.

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KEY DEFINITIONS

Artificial Narrow Intelligence (ANI)

AI capable within a specific domain, no meaningful transfer to other domains.

Large Language Model (LLM)

A computational system trained on human-generated text at scale. Capable of sophisticated language tasks and useful synthesis. Below the threshold of general cross-domain cognitive capability that warrants ASGI reclassification.

Artificial Simulated General Intelligence (ASGI)

AI demonstrating broad general cognitive capability across multiple domains whose knowledge rests on inherited simulation rather than any independently confirmable process — and possessing no internal mechanism to distinguish between the two. A small number of frontier large language models currently meet this definition.

Artificial General Intelligence (AGI)

Hypothetical AI with broad capability whose knowledge rests on a confirmed, independently verifiable foundation.

Artificial Superintelligence (ASI)

Hypothetical AI exceeding human capability across all domains.

The Citation Problem

The structural condition whereby an LLM's knowledge is entirely inherited from human-generated text and therefore only as reliable as the least trustworthy step in its provenance chain.

The Poisoning Attack Surface

The vulnerability of text-trained systems to deliberate or accidental corruption of their knowledge base through manipulation of training sources — compounding across generations as AI-generated content enters future training corpora.

The Capability Threshold

The point at which an LLM's demonstrated general cognitive capability across multiple domains warrants reclassification as ASGI. Defined by sustained cross-domain reasoning, novel synthesis, adaptive response to genuinely novel problem framings, and expert-level performance across multiple fields simultaneously.

Simulated Knowledge

Knowledge acquired through training on human-generated representations, inherited rather than independently confirmed.

Confirmed Knowledge

Knowledge whose basis can be independently examined, reproduced, and audited — where the confidence it carries reflects something more than the frequency with which it appeared in a training corpus.

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